Multimedia Signal Processing - 2nd Module

September, 3 2020

1. **Question (8 pts)** Consider three discrete sources X, Y and Z, each one emitting symbols from the alphabet $[a, b, c]$.
   (a) Compute the entropy bounds for the source X (i.e., minimum and maximum possible entropy).
   (b) Compute the maximum achievable joint entropy $H(X, Y)$.
   (c) Compute the maximum achievable joint entropy $H(X, Y, Z)$.

   You now observe the following sequence of symbols related to the source X

   $$x = \{a, b, a, c, a, a, b, c, a, c\}.$$

   (d) Estimate the pmf $p(X)$ of the source X by considering the sequence x as a representative example.
   (e) Compute the entropy $H(X)$.
   (f) Derive the Huffman code for the sequence x, showing a picture of the Huffman tree.

   You now observe the following sequence of symbols related to the source Y

   $$y = \{b, a, b, c, b, b, a, c, b, c\}.$$

   Consider the case of encoding pairs of symbols from X and Y (i.e., the i-th macro-symbol is the tuple (x_i, y_i) composed by the i-th element of x and the i-th element of y).
   (g) Estimate the joint pmf $p(X, Y)$ of the macro-symbols.
   (h) Compute the Huffman code for the macro-symbols (x_i, y_i) showing a picture of the Huffman tree.

2. **Question (7 pts)**
   Oral examination.

3. **Questions (8 pts - each answer can be either TRUE or FALSE - Incorrect answers give you -1 points, unanswered 0 points, correct answers 1 point - Provide a brief comment when in doubt)**

   - [T] [F] A scalar uniform quantizer and a Lloyd-Max quantizer provide the same solution for a uniformly distributed source.
   - [T] [F] Consider the coding process of a finite alphabet source X. If we allocate more than $H(X)$ bits per symbol we are surely applying a lossless coding scheme.

- [T] [F] In a closed-loop DPCM scheme, the distortion introduced on the signal is higher than the distortion introduced on the prediction residual.
- [T] [F] Considering a transform coding scheme, KLT transform is always preferable to DCT.
- [T] [F] Consider a speech coding DPCM scheme based on the combination of short-term and long-term predictor. The long-term predictor takes into account the periodicity introduces by the voice pitch.
- [T] [F] In perceptual audio coding, we typically allocate a minimum amount of bits greater than zero also to masked inaudible components.
- [T] [F] In image coding, changing the color space from R-G-B to Y-Cb-Cr may provide some advantages.
- [T] [F] In a video coding scheme, the decoder estimates the motion vectors to reconstruct the frames.

4. **Question (Matlab - 10pts)**

Consider the case of encoding the red color channel of an image comparing a PCM and a transform coding scheme based on 2D DCT. To do this, load the standard RGB image 'peppers.png' running the command `img = imread('peppers.png');` and follow these steps:

(a) Display the image in a figure.

(b) Save the red color channel as variable `R` and display it on screen.

(c) Encode and decode `R` using a uniform scalar quantizer (PCM) with the four quantization steps `Q` $\in$ {`10, 15, 20, 25`}.

(d) For each `Q` compute how many bit-per-pixel you need to save the encoded `R` (PCM case).

(e) For each `Q` compute the PSNR of the decoded `R` (PCM case).

(f) Encode and decode `R` using a 2D DCT transform coding scheme using 8x8 pixel blocks. Repeat the operation for the four quantization steps `Q` $\in$ {`10, 15, 20, 25`}.

(g) For each `Q` compute how many bit-per-pixel you need to save the encoded `R` (DCT case).

(h) For each `Q` compute the PSNR of the decoded `R` (DCT case).

(i) Draw the rate distortion curve for both coding schemes (i.e., PCM and DCT) using PSNR as distortion and bit-per-pixel as rate.